

ELEMENTARY CONSERVATION OF WATER

MODULE - 4

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Conserve Water to Save Life



Importance of conservation of water

- Freshwater is only available to us in limited quantity due to careless use of water and increasing level of pollution in water bodies.
- The shortage of water can lead to climate change.
- Water shortage can cause drought which then impacts animals and plants.
- We need water for our everyday activities from drinking to cleaning and washing.
- Ensuring a sustainable supply of fresh water for present and future generations.
- Promoting public health.
- Reducing energy consumption.
- Supporting ecological balance by minimizing the impacts of water scarcity on natural habitats.
- Preserving wildlife habitats.
- Minimizing the impacts of water pollution on the ecosystem.

Flow chart of steps to conserve water



Take shorter showers — five minutes or less is best.

Turn off the water while soaping hands and brushing teeth.



Turn off sink faucet while scrubbing dishes and pots.



Mulch around plants to hold water in the soil.



Get an Energy Star labeled washing machine.



Use a low flow showerhead.

Wash only full loads.



Install new toilets that use less than 1.6 gallons per flush.



Put faucet aerators on sink faucets.

Use a broom, not a hose, to clean driveways and walkways.



Water your yard and outdoor plants early or late in the day to reduce evaporation.

Use a shut-off nozzle on your hose.



Use plants that require less water.



Conservation of water in daily use:

At Home

- Turn off taps while brushing teeth or washing hands.
- Fix leaking taps and pipes to prevent water wastage.
- Use a bucket instead of a hose for washing vehicles.
- Take shorter showers and install water-saving showerheads.
- Use the washing machine and dishwasher only with full loads.

In the Garden

- Water plants early in the morning or late evening to minimize evaporation.
- Use a watering can instead of a hose.
- Collect rainwater in barrels to use for gardening.
- Plant drought-resistant and native plants that need less water.

General conservation:

- Use water-efficient appliances and fixtures.
- Recycle water where possible (e.g., reuse water from washing vegetables to water plants).
- Educate family and friends about the importance of saving water.



Household Water Use & Ways to Save



Install Water-Efficient Appliances & Fixtures

TOILETS

- Toilets account for approximately 24 percent of indoor water use. They also happen to be a major source of leaks and/or inefficiency.
- Over the course of your lifetime, you will likely flush the toilet nearly 140,000 times. A high-efficiency toilet can save you 4,000 gallons per year.
- A leaky toilet can waste 200 gallons of water per day.
- Test toilets for leaks. Add a few drops of food coloring or a dye tablet to the water in the tank, but do not flush the toilet. If the coloring appears in the bowl within a few minutes, the toilet has a leak that needs to be repaired (usually the flapper or rubber washer).

SHOWERS

- A water-efficient showerhead is one of the single most effective water-saving steps you can take inside your home.
- Take shorter showers.
- A full bathtub can use up to 70 gallons of water versus a 5-minute shower that uses as little as 10 gallons.

WASHING MACHINES

- High-efficiency washers use 35 to 55 percent less water and 50 percent less energy.
- They also require less detergent, rinse more thoroughly, are less abrasive on clothes, and can fit larger capacity loads in the same size drum.

DISHWASHERS

- High-efficiency dishwashers use 2.1 to 7 gallons of water per load.
- Replacing older, inefficient models can cut dishwasher water use in half.

SINKS

- Install faucet aerators on sinks for a simple, cost-effective way to save water. The faucet's efficiency can double without sacrificing performance.





Key Benefits of Water Recycling

1. Water Conservation: Recycling water reduces dependency on freshwater sources, ensuring that these resources are preserved for essential uses.

2. Environmental Protection: Treated wastewater can be safely released into natural water bodies, reducing pollution and supporting aquatic ecosystems.

3. Economic Efficiency: Recycled water can be used for agricultural irrigation, industrial processes, and landscaping, lowering costs associated with water extraction and transportation.

4. Climate Resilience: With climate change impacting water availability, recycling provides a reliable alternative during droughts and dry seasons.

Conserving Water At Home

10 ways to reduce your consumption, lower your bills, and help the environment

Upgrade Your Appliances



Turn Off The Tap



For every minute that a faucet runs two gallons of water are used.



Water Your Lawn Right



Water during sunrise or sunset to avoid losing water to evaporation.

Cut Your Grass Effectively

Keep your grass height at 2 inches to give it shade and help it retain moisture.



Use Grey Water

Grey water is relatively clean, used water from your bathroom sinks, showers, washing machine, & other appliances. It can be reused to water plants and gardens.

Use your dishwasher wisely

New dishwasher models use less water than hand washing and clean more thoroughly let you skip a pre-wash!



Compost

Garage disposals use a lot of water and they also add a considerable amount of solids to your septic tank. Compost instead!



Check For Leaks

A faucet leaking at one drip per second can waste 3,000 gallons a year



Take Shorter Showers

Reuse and Re-wear

Save water on laundry loads by re-wearing lightly worn clothes



Conservation of water in Agriculture:

- Agriculture uses ~ 70% of global freshwater — making it the largest consumer.
- Water scarcity is rising due to climate change, population growth, and inefficient practices.
- Conserving water helps ensure food security, environmental sustainability, and economic viability for farmers.

Factors affecting water scarcity in agriculture:

- Efficient Irrigation Systems
- Soil Management
- Crop Selection
- Runoff and Evaporation

Methods of water Conservation in Agriculture:

1. Drip Irrigation

- Delivers water slowly and directly to plant roots.
- Reduces evaporation and runoff significantly.



2. Mulching

- Applying straw, leaves, or plastic over soil.
- Keeps soil cool and moist, reducing evaporation.



3. Rainwater Harvesting

- Collect and store rain during the rainy season.
- Use it later for irrigation during dry spells.



4. Use of Drought-Resistant Crops

- Choose crops that need less water (e.g., millets, sorghum).
- Adapts farming to local water availability.

Drought Resistant Crops



5. Soil Moisture Monitoring

- Use sensors to check real-time soil moisture.
- Water only when necessary, preventing overwatering.



6. Contour Farming and Terracing

- Farming along the natural contours of land.
- Slows water runoff and increases water infiltration.

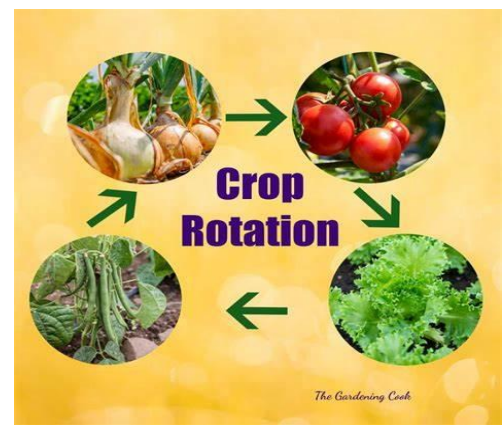


7. Scheduling Irrigation (Smart Watering)

- Water plants early in the morning or late in the evening.
- Minimizes water loss through evaporation.

8. Crop rotation

- Crop rotation is a farming method where different types of crops are planted in the same field across different seasons or years, instead of growing the same crop over and over.



9. Conservation Tillage

- Minimal plowing of land.
- Keeps organic matter and moisture locked in the soil.



10. Agroforestry

- Integrating trees into farming systems.
- Trees help shade soil, reduce evaporation, and improve water retention.



Methods of water Conservation in Industries:

1. Process Optimization

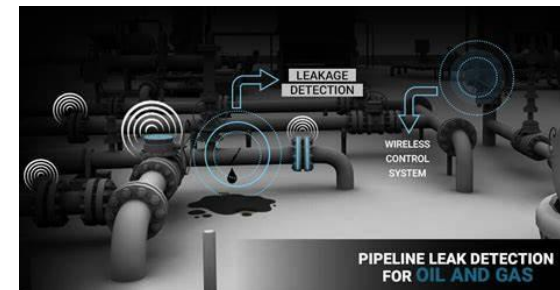
- **Recycle and reuse:** Implement closed-loop systems to reuse water within the process.
- **Water-efficient equipment:** Upgrade to machines that use less water.
- **Dry processes:** Where possible, replace wet processes with dry alternatives.

2. Water Audits

- Conduct regular water audits to identify usage patterns, leaks, and areas for improvement.
- Monitor flow meters and sub-meter usage to track consumption by department or activity.

3. Leak Detection and Repair

- Regularly inspect and promptly repair leaks in pipes, valves, and fittings.
- Use smart sensors for real-time leak detection.



Methods of water Conservation in Industries:

4. Cooling Systems

- Use air-cooled systems instead of water-cooled systems where feasible.
- Recycle cooling water through cooling towers with proper treatment.
- Install variable speed drives (VSDs) to control water flow more efficiently.

5. Wastewater Treatment and Reuse

- Treat and reuse greywater or process water in non-potable applications (e.g., cleaning, cooling).
- Install onsite wastewater treatment plants (ETPs/STPs).

6. Rainwater Harvesting

- Collect and store rainwater for use in non-critical applications.
- Reduce reliance on freshwater sources.

7. Employee Training and Awareness

- Educate employees on the importance of water conservation.
- Encourage suggestions for water-saving innovations.

Methods of water Conservation in Industries:

8. Automation and Smart Controls

- Use automated shut-off valves and sensors to prevent wastage.
- Monitor water usage through IoT-based water management systems.

9. Chemical Substitution

- Use chemicals that require less water for processing or cleaning.

Water Pollution Prevention Measures

1. Industrial Sector

- **Proper waste treatment:** Treat industrial effluents before discharge using Effluent Treatment Plants (ETPs).
- **Zero Liquid Discharge (ZLD):** Adopt systems where all wastewater is treated and reused within the facility.
- **Use of non-toxic materials:** Replace hazardous substances with environmentally friendly alternatives.
- **Spill prevention:** Install containment systems and train staff to handle chemicals safely.

2. Agricultural Sector

- **Controlled use of fertilizers and pesticides:** Apply chemicals judiciously and in the correct quantities.
- **Buffer zones and vegetation strips:** Plant trees or grasses along field edges to filter runoff.
- **Organic farming practices:** Reduce the use of synthetic inputs to lower pollution risks.
- **Proper animal waste management:** Store and treat livestock waste to prevent contamination.

Water Pollution Prevention Measures

3. Domestic Sector

- **Avoid dumping waste into drains:** Dispose of oils, paints, medicines, and chemicals properly.
- **Use eco-friendly cleaning products:** Choose biodegradable detergents and cleaners.
- **Install septic tanks correctly:** Ensure regular maintenance and prevent leakage.
- **Rainwater harvesting:** Reduce stormwater runoff that carries pollutants.